

Abdulmalik Abdulmawla

COMPUTATIONAL URBAN DESIGN · SPATIAL-ECONOMIC RESEARCH

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Spatial-economic researcher and computational urban designer. I analyze urban structures in relation to local value creation, service economies, and resilience — working with GIS, network analysis, agent-based simulation, and data-driven methods (Python, R, C#) to empirically link spatial configuration with economic dynamics, and building the research tools that make that analysis reproducible.

EDUCATION

2018 – present

PhD candidate — in progress · Computer Science in Architecture & Urbanism (InfAR), Bauhaus-Universität Weimar

Thesis: *“Spatial dependency of urban retail: detailed methods for positioning street-level activities”* — supervisor: Prof. Dr. Reinhard König

2013

Master of Arts in Architecture · Hochschule Anhalt, Dessau-Roßlau, Germany

2007

BSc Architecture & Engineering · Faculty of Engineering, Mansoura University, Egypt

EXPERIENCE

Dec 2023 – present

Research Associate — Chair of Transport System Planning

Bauhaus-Universität Weimar · Bauhaus Mobility / European Digital Innovation Hub Thuringia (EDIH.TH)

Collection and analysis of open mobility data; interactive data-exploration platforms; data-science consulting for public institutions, incl. training public administrations on European open-mobility-data standards and Mobilithek publishing; supervision of student theses (Digital Transport Research Lab).

2016 – 2023

Research Associate — Computer Science in Architecture (InfAR/InfAU)

Bauhaus-Universität Weimar

Teaching computational methods for urban planning and analysis; development of design and research tools (DeCodingSpaces Toolbox, mineR); master's-thesis supervision; conference research (Space Syntax Symposium, eCAADe) and support of academic publications.

2017 – 2019

Project Coordinator — Discovering City (DAAD exchange)

Bauhaus-Universität Weimar ↔ German-Jordanian University, Amman

Wrote the project proposal, managed the finances, organized the teaching and workshop programme.

2016 – 2020

Contributing Author — DeCodingSpaces Toolbox

DecodingSpaces GbR, Weimar

C# development for a Grasshopper toolbox for urban analysis; testing, documentation, consulting.

2014 – 2016

Research Associate — Teaching & Computational Design

Arab Academy for Science, Technology & Maritime Transport (AASTMT), Cairo

Computational and parametric design (Grasshopper, Processing, Arduino) in a RIBA Part I certified programme.

2012 – 2016

Co-founder & Architect — Gozour Studio

Mansoura, Egypt

Residential, education and competition projects (incl. Lusail Underpass Competition, Qatar).

2005 – 2011

Junior & Freelance Architect — Egypt & UAE

EAI Group, Abu Dhabi · Invert Studios, Cairo · Al Turath, Abu Dhabi

Early architectural practice: design, CAD, 3D modelling and visualization, client representation.

SELECTED PUBLICATIONS

- 2022** — Abdulmawla, A., Bielik, M., Schneider, S., & König, R. Mapping active frontages: a method for linking street-level activities to their surrounding urban forms — a case study of Weimar, Germany. *13th International Space Syntax Symposium (SSS13)*, Bergen.
- 2022** — Abdulmawla, A., Schneider, S., König, R., Bielik, M., & Fuchkina, E. Parametric urban data structuring and spatial query. *40th eCAADe*.
- 2019** — Bielik, M., König, R., Fuchkina, E., Schneider, S., & Abdulmawla, A. Evolving configurational properties: simulating multiplier effects between land use and movement patterns. *12th International Space Syntax Symposium (SSS12)*.
- 2018** — Abdulmawla, A., Schneider, S., Bielik, M., & König, R. Integrated data analysis for parametric design environments — mineR: a Grasshopper plugin based on R. *36th eCAADe*.
- 2017** — König, R., Bielik, M., Knecht, K., Abdulmawla, A., & Fuchkina, E. New methods for urban analysis and simulation with Grasshopper — the DeCodingSpaces-Toolbox. *CAAD Futures 2017*, Istanbul.
- 2017** — Schneider, S., Abdulmawla, A., & Donath, D. The effect of the street network on movement patterns and land use in small cities. *11th International Space Syntax Symposium*, Lisbon.
- 2017** — Denmark, M., Schneider, S., König, R., Abdulmawla, A., & Donath, D. Towards a modular design strategy for urban masterplanning. *eCAADe 2017*, Rome.

Full list: researchgate.net/profile/Abdulmalik-Abdulmawla

TEACHING & WORKSHOPS

University seminars: Algorithmic Architecture · Parametric Urban Design & Analysis · SYNCITY I & II · Circular Urbanism · Digital Traffic Simulation Lab · Open Mobility Data · Rent-a-Data-Scientist.

Workshops (organizer): Discovering Cities (GJU Amman, 2018) · Urban Analysis, Synthesis & Exploration with Grasshopper (eCAADe 2018, Łódź) · Rural-to-Urban Transformation (Emerging City Lab, Addis Ababa, 2016) · Simulating Agent-Based Systems using Processing (Cairo, 2015).

SKILLS

Programming: Python, C#, R, Grasshopper, Java/Processing, vvvv, Arduino

Spatial & analytical: GIS, network analysis, agent-based simulation, space syntax, spatial data engineering

Modelling & media: Rhino, AutoCAD, Revit, 3ds Max, Photoshop, Illustrator, InDesign

Languages: Arabic (native) · English (fluent) · German (very good)

RECOGNITION

- 2026** — Ten-year service recognition at the Bauhaus-Universität Weimar, presented by the University President.